

Twin Pregnancy 04.

Monochorionic Disease

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode four. Monochorionic disease.

This episode is based mainly on Gabbe chapter 32, with support from the Israeli twin follow-up paper, the fetal echocardiography position paper, the amniotic fluid position paper, and the fetal reduction position paper.

Opening frame

Let us begin with the deepest idea.

Monochorionic twins are not high risk merely because there are two fetuses. They are high risk because there is one placenta with vascular connections between two fetal circulations.

That single fact explains the major diseases: twin-to-twin transfusion syndrome, or TTTS, selective fetal growth restriction, or sFGR, twin anemia polycythemia sequence, or TAPS, twin reversed arterial perfusion sequence, or TRAP, and injury after single fetal demise.

Start with the placenta

Start with the placenta.

In monochorionic twins, vascular anastomoses are common. Some are superficial and bidirectional, especially arterioarterial and venovenous connections. Others are deep arteriovenous connections through a shared cotyledon, where arterial blood from one twin drains into the venous side of the other.

Balanced vascular sharing may be tolerated. Unbalanced sharing becomes disease.

TTTS is the classic example. It occurs only in monochorionic pregnancies, most often monochorionic diamniotic pregnancies. The donor twin sends more blood than it receives. The recipient receives more than it sends.

The donor becomes hypovolemic. Renal perfusion falls. Urine production falls. Amniotic fluid falls. The donor may become growth restricted and appears stuck against the uterine wall when oligohydramnios is severe.

The recipient becomes hypervolemic. It urinates more, develops polyhydramnios, and may develop cardiac strain, ventricular hypertrophy, valve regurgitation, hydrops, or

fetal death.

The ultrasound diagnosis is not simply size difference. The classic diagnostic logic is monochorionic diamniotic twins with oligohydramnios in one sac and polyhydramnios in the other. Gabbe uses deepest vertical pocket less than 2 centimeters in the donor sac and greater than 8 centimeters in the recipient sac.

Quintero staging is useful, but it should not be worshipped. The donor bladder, Dopplers, hydrops, and demise matter, but disease may not progress stepwise. A fetus can deteriorate without politely passing through each stage. Also, Quintero staging can underrepresent recipient cardiac dysfunction.

Why do we scan monochorionic diamniotic twins every two weeks beginning around 16 weeks?

Because TTTS often presents between 15 and 26 weeks, and longer ultrasound intervals are associated with more advanced disease at diagnosis. Surveillance is not about collecting pretty growth scans. It is about detecting a hemodynamic disease while fetal therapy is still possible.

Treatment depends on gestational age and stage, but the conceptual treatment is clear. Amnioreduction removes the consequence, which is polyhydramnios. Fetoscopic laser photocoagulation treats the cause, which is the placental vascular anastomoses. That is why laser is considered optimal therapy for stage two to four disease before 26 weeks in many expert frameworks.

Now selective fetal growth restriction

Now selective fetal growth restriction.

sFGR means one twin is growth restricted while the co-twin is not, especially in monochorionic twins. The pathophysiology is often unequal placental territory, abnormal cord insertion, or both. But in monochorionic twins, umbilical artery Doppler does not mean exactly what it means in a singleton. The waveform reflects placental resistance plus the influence of intertwin vascular communications.

This is why residents should be careful. In a singleton, absent or reversed end-diastolic flow usually signals severe placental insufficiency and limited latency. In monochorionic sFGR, abnormal Dopplers can behave differently because arterioarterial connections may intermittently modify flow. The pregnancy can be unstable in a way that singleton FGR algorithms do not fully capture.

Now TAPS

Now TAPS.

TAPS is chronic hemoglobin discordance without the amniotic fluid discordance of classic TTTS. Think of very small arteriovenous connections allowing slow net transfusion. One twin becomes anemic, the other polycythemic. The ultrasound clue is middle cerebral artery peak systolic velocity. The anemic twin has high velocity, classically 1.5 multiples of the median or more. The polycythemic twin has low velocity.

The key contrast is this. TTTS is fluid and volume imbalance. TAPS is red-cell imbalance without meeting TTTS fluid criteria.

Now TRAP sequence

Now TRAP sequence.

Twin reversed arterial perfusion sequence occurs in monochorionic twins when one twin, the acardiac twin, is perfused by the pump twin through abnormal vascular connections. The acardiac twin is not viable. The pump twin is at risk because it must support its own circulation and the abnormal vascular mass of the co-twin. The clinical problem is not saving the acardiac twin. It is protecting the pump twin from high-output failure and preterm birth.

The fetal echocardiography position paper lists monochorionic multiple pregnancy and acardiac twin among indications for fetal echocardiography. That makes sense. These are cardiovascular diseases of placentation.

Now monoamniotic twins

Now monoamniotic twins.

Monoamniotic twins share one chorion and one amniotic sac. They carry the monochorionic vascular risks plus the unique risk of cord entanglement. Some degree of cord entanglement is common, but not all entanglement predicts immediate demise. The difficulty is that monitoring may not reliably prevent all cord accidents.

This uncertainty explains the usual strategy: very close surveillance, often hospitalization in some protocols, corticosteroids, and planned cesarean delivery around 32 to 34 weeks, with local protocol leaning 32 to 33 weeks.

Now single fetal demise

Now single fetal demise.

In dichorionic twins, demise of one fetus is tragic, but the surviving fetus does not share direct placental vessels with the demised co-twin. Early delivery is not automatically required.

In monochorionic twins, demise of one fetus can acutely injure the survivor. Blood can shift through vascular anastomoses into the low-pressure circulation of the demised twin. The surviving twin can suffer hypotension, anemia, ischemic brain injury, or death. Immediate delivery does not reliably prevent damage, because injury may occur at the moment of demise. The Israeli paper recommends MCA Doppler after diagnosis to assess anemia and brain imaging with ultrasound or MRI to look for injury.

Pause and summarize.

Monochorionic disease is vascular disease. TTTS is unbalanced volume and fluid. sFGR is unequal placental share plus complex Doppler interpretation. TAPS is slow red-cell transfusion. TRAP is pump-twin cardiovascular burden. Single demise is dangerous because the surviving twin may acutely exsanguinate into the co-twin circulation.

Now final synthesis

Now final synthesis.

Key point

First, monochorionicity means one placenta and vascular connections.

Second, every major monochorionic complication is easier to understand if you ask what is being shared: blood volume, red cells, placental territory, cardiac work, amniotic space, or pressure after demise.

Third, surveillance every two weeks from about 16 weeks in monochorionic diamniotic twins exists because disease can develop quickly and treatment is time-sensitive.

Fourth, monochorionic complications should trigger referral to a center that can provide high-level fetal assessment and fetal therapy when appropriate.

End of episode four.